

$$b) A = \{1, 2, 3, 4\}, B = \{3, 4, 5, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\} \quad A \cap B = \{3, 4\}$$

$$|A \cup B| = 6 \quad |A \cap B| = 2 \quad J(A, B) = \frac{2}{6} = \frac{1}{3} \quad d_J(A, B) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$c) A = \{1, 2, 3, 4, 5, 6\}, B = \{1, 2, 3, 4, 5, 6\}$$

$$A \cup B = A \cap B = A = B$$

$$|A \cup B| = |A \cap B| = 6 \quad J(A, B) = \frac{6}{6} = 1 \quad d_J(A, B) = 1 - 1 = 0$$

$$d) A = \{1\}, B = \{1, 2, 3, 4, 5, 6\}$$

$$A \cup B = B \quad A \cap B = A$$

$$|A \cup B| = 6 \quad |A \cap B| = 1 \quad J(A, B) = \frac{1}{6} \quad d_J(A, B) = 1 - \frac{1}{6} = \frac{5}{6}$$

72) Prove that each of the properties in parts (a)-(d) holds whenever A and B are finite sets.

$$a) J(A, A) = 1 \text{ and } d_J(A, A) = 0$$

$$\text{Solution: } J(A, A) = \frac{|A \cap A|}{|A \cup A|} = \frac{|A|}{|A|} \quad (\text{1st Idempotent Law})$$

$$= \frac{|A|}{|A|} \quad (\text{2nd Idempotent Law}) = 1$$

$$d_J(A, A) = 1 - 1 = 0 \quad \square$$

$$b) J(A, B) = J(B, A) \text{ and } d_J(A, B) = d_J(B, A)$$

$$\text{Solution: } J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|B \cap A|}{|B \cup A|} \quad (\text{2nd Commutative Law})$$

$$= \frac{|B \cap A|}{|B \cup A|} \quad (\text{1st Commutative Law})$$

$$d_J(A, B) = 1 - J(A, B) = 1 - J(B, A) = d_J(B, A) \quad \square$$

$$c) J(A, B) = 1 \text{ and } d_J(A, B) = 0 \text{ if and only if } A = B.$$

$$\text{Solution: } J(A, B) = 1 \Rightarrow \frac{|A \cap B|}{|A \cup B|} = 1 \Rightarrow |A \cap B| = |A \cup B|$$

By Prob 16(a), $A \cap B \subseteq A$. By Prob 16(b), $A \subseteq A \cup B$.

By Prob 19 from Ex-2.1, $A \cap B \subseteq A \cup B$. By Defn. 3 from Sec-2.1

$$\text{if } x \in A \cap B, \text{ then } x \in A \cup B. \quad \therefore |A \cap B| \leq |A \cup B|$$